

SOUSANNAH ABDALLA

AI Engineer | Full Stack Developer | Technical Lead
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Professional Summary

AI Engineer and Technical Lead with 2+ years of experience building production AI systems and full-stack applications. Hands-on expertise in Python, LLM orchestration, prompt engineering, RAG, model fine-tuning and evaluation, speech AI, Arabic OCR, computer vision, and REST API integration. Founded Odenta, an AI-powered dental platform deployed as the official system at Alamein International University, and published an IEEE ICUAS 2025 paper.

Technical Skills

AI and Machine Learning: Python, PyTorch, TensorFlow, Keras, Scikit-learn, Machine Learning, Deep Learning, Neural Networks, Generative AI, Large Language Models, Fine-Tuning, RAG, Prompt Engineering, AI Agents, FWA Detection, Information Retrieval, Model Evaluation, Recommendation Systems, Ranking

Voice AI and Computer Vision: Conversational AI, Speech-to-Text, Text-to-Speech, Speech Emotion Recognition, OpenCV, YOLO, MediaPipe, OCR, Arabic OCR, Document AI, Object Detection, Image Classification, Pose Estimation

Full Stack and Mobile: JavaScript, React, React Native, iOS and Android Development, Tailwind CSS, Node.js, Express.js, MongoDB, MERN Stack, Flask, FastAPI, REST APIs, SQL, Authentication, Role-Based Access Control, Data Processing, Debugging

Cloud and Engineering: Docker, Git, AWS, GCP, NVIDIA DGX, Model Serving, GPU Monitoring, API Integration, Agile Delivery, System Architecture, Technical Leadership

PROFESSIONAL EXPERIENCE

Healthplans.AI

Sep 2025 - Present

AI Engineer and Technical Lead

USA

- Led the architecture, implementation, and UAT of a claim-status voice agent that captured provider information, authenticated member and claim data before disclosure, applied information retrieval to claim records, and provided human-like explanations for paid, denied, or pending claims.
- Owned the end-to-end voice workflow across speech-to-text, text-to-speech, LLM orchestration, API integrations, call-session management, and secure healthcare data handling.
- Led a suite of Fraud, Waste, and Abuse agents that processed provider and claims data, detected multiple fraud typologies, and generated structured outputs for investigation workflows.

Odenta

2025 - Present

Founder and Lead Engineer

Egypt

- Founded and launched an AI-powered dental management platform used by universities and clinics; delivered its successful Summer 2025 trial and deployment as Alamein International University's official dental system.
- Architected and developed the MERN platform for 9,000+ patient records, including appointments, tooth charts, X-rays, analytics, reviews, messaging, invoicing, and role-based workflows for clinical and administrative users.
- Designed the new Odenta Voice architecture, enabling clinicians to interact with schedules, patient history, and clinical workflows through secure, human-like voice conversations.
- Developed the Odenta mobile application with React Native for both iOS and Android, extending core clinical and practice management workflows to mobile users.
- Own product strategy, technical architecture, deployment, stakeholder communication, and delivery priorities across Odenta's web, mobile, and AI products.

Maxbit LLC

Oct 2025 - Jan 2026

AI Engineer

Egypt

- Fine-tuned and evaluated DeepSeek, Qwen2.5, and vision-language models to improve Arabic OCR for national IDs, passports, and other identity documents.
- Developed real-time speech emotion recognition for live audio and built a DGX monitoring dashboard for GPU, memory, utilization, and workload visibility.

Alamein International University

Sep 2025 - Present

Teaching Assistant

Egypt

- Teach AI and computer science labs, guide semester projects, assess technical work, and mentor students in applied engineering problem-solving.

Fiverr

Jan 2025 - Present

Freelance AI Engineer and Full Stack Developer

Remote

- Deliver AI-powered applications, machine-learning integrations, REST APIs, and MERN solutions for international clients while maintaining a 5.0/5.0 rating.

Zudu AI

Mar 2025 - May 2025

AI Engineer

Remote

- Developed a human-like voice agent for customer calls, improving speech recognition, dialogue flow, response orchestration, and automated support interactions.

Digital Egypt Pioneers Initiative

Apr 2024 - Oct 2024

React Developer Intern

Egypt

- Built responsive full-stack applications using React, Node.js, JavaScript, REST APIs, and reusable UI components for digital transformation projects.

University of Louisville

AI Engineer Intern

Sep 2023 - Nov 2023

USA

- Designed deep-learning models for non-invasive age-related macular degeneration detection and severity grading with teams across the USA, Egypt, and Spain.

LARRI, University of Louisville

Computer Vision Intern

Jul 2023 - Sep 2023

USA

- Built real-time hand-gesture recognition with OpenCV, MediaPipe, and YOLO, then integrated gesture-based UAV control using DJI Tello and AirSim.

PROJECTS

Viatryon

- Built a MERN and computer-vision virtual try-on platform for watches and bracelets, implementing real-time wrist detection, scaling, and rotation.

ASL Learning Platform and Human-Mimicking Robot

- Developed an AI-powered platform for live gesture-to-text translation and interactive assessments, using pose estimation and keypoint-based control to guide a human-mimicking robot.

EDUCATION

Alamein International University

Bachelor of Engineering, Artificial Intelligence

Sep 2021 - Sep 2025

Egypt

- **GPA:** 3.86/4.0

- **Coursework:** Machine Learning, Deep Learning, NLP, Computer Vision, Database Systems, Algorithms, Digital Signal Processing

Publication

- Abdalla, S..Comparative Analysis of Hand Gesture Detection Methods for Edge-Assisted UAV Control..*IEEE International Conference on Unmanned Aircraft Systems, 2025.* [arXiv:2505.17303](https://arxiv.org/abs/2505.17303)

Leadership

- * **Technical Leadership:** Lead architecture, engineering execution, UAT, and delivery for enterprise healthcare voice-agent and FWA initiatives at Healthplans.AI.
- * **Product Leadership:** Founded Odenta and own its product roadmap, web and mobile architecture, voice-AI integrations, production deployment, and university and clinic stakeholder delivery.
- * **Mentoring:** Teach university AI and computer science labs and coach competitive programming students in C++ and problem solving.

Languages

- Arabic(Native)
- English(Professional Working Proficiency)